

Notes: Tomunen (RFS, 2026)

Failure to Share Natural Disaster Risk

Contents

1	Executive summary	3
2	Institutional background: cat bonds in one page	3
2.1	What is a cat bond?	3
2.2	Key contract objects in the paper	3
2.3	Why the market structure helps identification	4
3	Theory model (all equations and the core derivations)	4
3.1	Setup: assets and payoffs (Equations (1)–(4))	4
3.2	Households: identification restriction (Equation (5))	4
3.3	Intermediaries: segmented access and outside capital constraint (Equation (6))	4
3.4	Manager objective: mean–variance preferences (Equation (7))	5
3.5	Portfolio wealth and market clearing (Equation (8))	5
3.6	Equilibrium pricing: CAPM-like relation (Equation (9))	5
3.7	Aggregate premium and the role of α (Equation (10))	5
4	Data and descriptive statistics (Table 1 and Figure 2)	5
4.1	How to read Table 1	5
4.2	Figure 2: specialists are “the” relevant investors, and macro comovement is small	6
5	Empirical implementation: measuring $E_t(R_i^e)$ and β_i	6
5.1	Expected excess returns	6
5.2	Modeled bond-state returns (Equation (11))	7
5.3	From simulations to cat-market betas	7
5.4	Determinants of simulated betas (Equations (12)–(13))	7
5.5	Interpreting Table 2: what drives beta dispersion	7
6	Main evidence: “failure to share” disaster risk	8
6.1	Cross-sectional pricing test (Equation (14) and Table 3)	8
6.2	Figure 1: security-level fit of the intermediary model	9
6.3	Characteristics vs factor exposure (Table 4)	9
7	Time-series and pooled evidence on intermediary capital	9
7.1	CRRA mapping (Equation (15)) and price of risk (Equation (16))	9
7.2	Figure 3: aggregate premium weakens as capital deepens	10
7.3	Pooled restriction and estimation (Equations (17)–(18); Table 5)	10

8	Testing the exclusion restriction: disasters and macro aggregates	11
8.1	Disaster damage measure (Equation (19))	11
8.2	Local projections (Equation (20)) and Figure 4	12
9	Ruling out alternative explanations	12
9.1	Liquidity (Table 6)	12
9.2	Climate change concerns (Figure 5)	13
9.3	Other explanations	13
10	Short summary	13
11	Conclusion	13
11.1	Summary	13
11.2	Why “failure to share” shows up as a premium	14
11.3	Key quantitative takeaways	15
11.4	Takeaway	15

1 Executive summary

- **Question.** Do asset prices reflect the marginal utility of *financial intermediaries* rather than households in a setting where omitted macro risk factors are unlikely?
- **Setting.** Catastrophe bonds (cat bonds): cash flows depend on natural disasters (hurricanes, earthquakes, etc.). These events are largely orthogonal to aggregate consumption/wealth at the macro scale, making identification cleaner than in traditional asset classes.
- **Main empirical result.** A one-factor intermediary model explains a large share of *security-level* expected return variation: about **71%** (illustrated in Figure 1).
- **Identification logic.** Under the maintained assumption that disasters are independent of households' marginal rate of substitution (MRS), a significant price of catastrophe-market risk cannot be reinterpreted as compensation for unobserved macro risks.
- **Time-series implication.** The *aggregate* premium on disaster risk decreases over time and becomes less sensitive to disasters when intermediaries' access to outside capital improves (Figure 3).
- **Additional evidence.** Macroeconomic impulse responses to natural disasters are near zero in a long panel of developed countries (Figure 4), supporting the key exclusion restriction.

Big-picture interpretation: Even if natural disaster risk is diversifiable for the economy as a whole, it can be *priced* if the marginal investors are specialized intermediaries who end up bearing concentrated disaster exposure because of segmentation and capital-raising frictions.

2 Institutional background: cat bonds in one page

2.1 What is a cat bond?

A cat bond is a security issued (often via an SPV) to transfer catastrophe risk from an insurer/reinsurer to capital-market investors. Investors receive coupons (reference rate + spread/discount margin) as long as no qualifying catastrophe event triggers principal losses. If a qualifying event occurs and losses exceed contract thresholds, principal can be partially or fully written down.

2.2 Key contract objects in the paper

For each bond i and time t :

- **Attachment probability** $ap_{i,t}$: probability the bond loses its first dollar of principal.
- **Exhaustion probability** $ep_{i,t}$: probability of full principal loss.
- **Expected loss** $el_{i,t}$: actuarial expected principal loss (annualized).
- **Discount margin** $dm_{i,t}$: market-implied spread-like object that prices the bond when not at par.
- **Expected excess return** $E_t(R_{i,t+1}^e)$: measured (roughly) as $dm_{i,t} - el_{i,t}$ after the paper's adjustments.

2.3 Why the market structure helps identification

Participation is effectively limited to sophisticated institutions and specialized ILS fund managers. The market is small relative to aggregate wealth, and disaster shocks are idiosyncratic at the macro level. Hence, if catastrophe risk is priced, it is natural to interpret pricing through *intermediary* marginal utility rather than household MRS.

3 Theory model (all equations and the core derivations)

3.1 Setup: assets and payoffs (Equations (1)–(4))

Two-period economy $t \in \{0, 1\}$ with a risk-free asset (rate r) and N cat bonds. Each cat bond is issued at par $P_{i,0} = 1$, supply θ_i , and pays coupon $C_i = r + s_i$ at maturity if principal remains.

Let $\xi_i \sim U(0, 1)$ represent catastrophe severity for bond i (smaller ξ means worse outcome). Let $\bar{x}_i$ be the attachment probability and x_i the exhaustion probability (so $0 \leq x_i \leq \bar{x}_i \leq 1$). Principal at maturity is piecewise:

$$P_{i,1}(\xi_i) = \begin{cases} 1, & \xi_i > \bar{x}_i, \\ F_i(\xi_i), & x_i \leq \xi_i \leq \bar{x}_i, \\ 0, & \xi_i < x_i, \end{cases} \quad (1)$$

with $F_i(x_i) = 0$, $F_i(\bar{x}_i) = 1$, and $F'_i(\xi) \geq 0$.

Define expected loss $el_i \equiv 1 - \mathbb{E}_0[P_{i,1}(\xi_i)]$. Total payoff is

$$X_{i,1} = P_{i,1}(\xi_i) + C_i. \quad (2)$$

Return is

$$R_i(\xi_i) = \frac{P_{i,1}(\xi_i) + C_i - P_{i,0}}{P_{i,0}} = P_{i,1}(\xi_i) + r + s_i - 1, \quad (3)$$

and expected excess return satisfies

$$\mathbb{E}_0(R_i^e) = s_i - el_i. \quad (4)$$

3.2 Households: identification restriction (Equation (5))

Let M_1 denote the representative household's MRS between $t = 0$ and $t = 1$. The key identifying assumption is:

$$M_1 \perp \xi_i \quad \forall i. \quad (5)$$

Interpretation. Cat-bond cash flows are orthogonal to household MRS, so cat risk should be *unpriced* in a frictionless representative-agent macro model. If cat risk is priced, it must come from *who bears the risk* and *what frictions prevent full sharing*.

3.3 Intermediaries: segmented access and outside capital constraint (Equation (6))

Specialist fund managers intermediate cat-bond risk to outsiders. Outsiders cannot directly buy cat bonds; they invest via these funds. To discipline managers, outsiders require “skin in the game”: if $w_{m,0}$ is manager wealth and w_0 is total fund size,

$$w_{m,0} \geq \alpha w_0. \quad (6)$$

Interpretation. Smaller α means the manager can lever outside capital more easily; larger α is a tighter equity constraint.

3.4 Manager objective: mean–variance preferences (Equation (7))

Managers are risk averse with mean–variance utility over terminal wealth:

$$U(w_{m,1}) = \mathbb{E}_0[w_{m,1}] - \frac{\rho}{2} \text{Var}_0[w_{m,1}]. \quad (7)$$

3.5 Portfolio wealth and market clearing (Equation (8))

Let q_i be the quantity of bond i held by the representative manager. Given the outside capital constraint, the manager can operate a fund of total size $w_{m,0}/\alpha$. If the remainder is invested in the risk-free asset, date-1 portfolio value is:

$$w_1 = \sum_i q_i (P_{i,1}(\xi_i) + C_i) + \left(\frac{w_{m,0}}{\alpha} - \sum_i q_i \right) (1 + r). \quad (8)$$

3.6 Equilibrium pricing: CAPM-like relation (Equation (9))

Under the paper’s equilibrium (FOC + market clearing), expected excess returns satisfy:

$$\mathbb{E}_0(R_i^e) = \beta_i \mathbb{E}_0(R_{\text{cat}}^e), \quad (9)$$

where R_{cat}^e is the excess return on the *cat market factor portfolio* (the aggregate cat-bond portfolio), and

$$\beta_i \equiv \frac{\text{Cov}_0(R_i^e, R_{\text{cat}}^e)}{\text{Var}_0(R_{\text{cat}}^e)}.$$

Economic logic. Even though catastrophe risk is diversifiable for the economy, it becomes *systematic* for specialists who must hold concentrated cat exposure in equilibrium.

3.7 Aggregate premium and the role of α (Equation (10))

The model predicts:

$$\mathbb{E}_0(R_{\text{cat}}^e) = \alpha a \rho \text{Var}_0(R_{\text{cat}}^e), \quad (10)$$

where a is the average specialist’s position in the cat market.

- If $\alpha = 0$ (no “skin-in-the-game” friction), the premium vanishes and risk is efficiently shared.
- If $\alpha > 0$, specialists cannot scale outside capital freely, so they bear too much cat risk and the premium is positive.

4 Data and descriptive statistics (Table 1 and Figure 2)

4.1 How to read Table 1

Table 1 is the “map” of the market and the paper’s core objects.

Panel A (primary market bond characteristics). The typical issuance is about \$130M with a term around 3 years. The mean spread is about 7.3% while the mean actuarial expected loss is only 2.3%, so the mean expected excess return $E(R^e)$ is about 5.2%. This is the paper’s first descriptive indication of a *large premium* on catastrophe risk.

Table 1
Summary statistics

A. Bond characteristics (primary market)								
Variable	N	Mean	SD	Min	25%	50%	75%	Max
Size (\$ million)	675	130.8	119.6	1.8	50.0	100.0	175.0	1500.0
Term (months)	675	36.9	13.1	5.0	35.4	36.3	47.6	120.0
Spread (%)	658	7.3	5.1	0.7	4.0	6.0	9.3	49.2
ap (%)	669	3.1	3.2	0.0	1.1	2.0	4.2	23.2
el (%)	669	2.3	2.3	0.0	0.8	1.4	3.1	15.8
ep (%)	669	1.7	1.8	0.0	0.6	1.1	2.3	12.0
$E(R^e)$ (%)	653	5.2	3.7	0.6	2.8	4.3	6.3	43.0

B. Spreads by expected loss quintile (primary market)								
Quintile	N	Mean	SD	Min	25%	50%	75%	Max
1	134	3.8	1.8	0.7	2.3	3.6	4.9	10.2
2	134	5.0	2.1	1.8	3.2	4.5	6.4	12.5
3	130	6.4	2.8	2.9	4.5	5.8	7.5	20.5
4	133	8.5	3.4	3.0	6.0	7.5	10.8	17.8
5	122	13.7	6.7	5.9	9.5	12.4	15.2	49.2

C. Secondary market								
Variable	N	Mean	SD	Min	25%	50%	75%	Max
Turnover (%)	5,969	35.8	79.1	0.0	0.0	8.0	37.9	2000.0
dm^{sheet} (%)	6,538	6.6	4.8	0.0	3.4	5.3	8.1	39.5
dm^{trace} (%)	3,994	6.7	4.6	-0.3	3.5	5.4	8.3	36.9

D. Simulated $\hat{\beta}$								
Variable	N	Mean	SD	Min	25%	50%	75%	Max
$\hat{\beta}$	4,548	1.14	0.82	-0.00	0.49	1.11	1.63	6.39

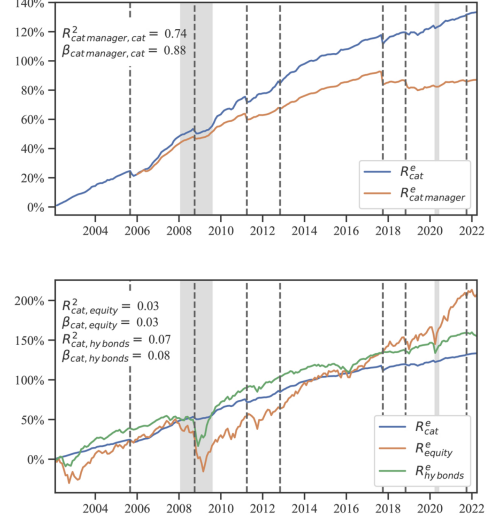


Figure 2
Cumulative excess returns on selected indices

Panel B (spreads by expected-loss quintile). Spreads rise strongly with expected loss: the mean spread moves from about 3.8% in the lowest expected-loss quintile to about 13.7% in the highest. This is consistent with a positive risk–return relation, but it does *not* by itself tell us whether pricing is driven by macro risks, intermediary constraints, or other frictions.

Panel C (secondary market liquidity and pricing objects). Trading is sparse: turnover has a high mean but a low median, and a large mass of zero-trading quarters. Discount margin measures from quotes and TRACE trades (dm^{sheet} and dm^{trace}) are centered around 6–7%. This matters because the empirical expected excess return in the secondary market is built from $dm_{i,t} - el_{i,t}$.

Panel D (simulated betas). Simulated cat-market betas have mean about 1.14 and substantial dispersion (up to about 6.4). This is essential: without cross-sectional beta heterogeneity, the pricing test cannot have power.

4.2 Figure 2: specialists are “the” relevant investors, and macro comovement is small

Figure 2 complements Table 1 by showing: (i) the cat-market factor is tightly linked to specialist manager returns, and (ii) cat-market returns do not line up with equity and HY corporate bond markets. This is the key descriptive support for the identification strategy.

5 Empirical implementation: measuring $E_t(R_i^e)$ and β_i

5.1 Expected excess returns

At issuance, Equation (4) implies $E_0(R_i^e) = s_i - el_i$. In secondary markets, the paper replaces the quoted spread by the **discount margin** $dm_{i,t}$ to incorporate capital gains/losses. The empirical object used in the main plots and regressions is:

$$E_t(R_{i,t+1}^e) \approx dm_{i,t} - el_{i,t},$$

after adjustments that improve the measurement of conditional loss probabilities between reset dates.

5.2 Modeled bond-state returns (Equation (11))

To estimate catastrophe-factor betas without relying on a short, skewed realized-return sample, the paper simulates bond returns. After adjusting for (i) bond-specific benchmark rate $ref_{i,t}$ and (ii) coupon accrual conditional on principal survival, modeled excess return in state ξ_i is:

$$R_{i,t+1}^e(\xi_i) = (1 + ref_{i,t} + dm_{i,t})P_i(\xi_i) - (1 + r_t). \quad (11)$$

5.3 From simulations to cat-market betas

- **Peril categories.** Each bond is assigned to a peril category (e.g., North America hurricane, Japan earthquake, etc.).
- **Correlation structure.** Disasters are assumed independent across peril categories; within a peril category, bonds share a common shock.
- **Monte Carlo.** For each time t and peril category, simulate 500,000 draws of disaster outcomes, compute bond excess returns using (11).
- **Factor portfolio.** The cat factor return is the par-value-weighted return across outstanding cat bonds.
- **Beta estimation.** $\hat{\beta}_{i,t}$ is the slope from a regression of simulated $R_{i,t+1}^e(\xi)$ on simulated $R_{cat,t+1}^e(\xi)$.

5.4 Determinants of simulated betas (Equations (12)–(13))

Simulated betas are a nonlinear function of bond characteristics and market composition:

$$\hat{\beta}_{i,t} = f(api_{i,t}, eli_{i,t}, epi_{i,t}, dm_{i,t}, ref_{i,t}, Peril_t, Size_t), \quad (12)$$

where $Peril_t$ and $Size_t$ summarize the cross-section of outstanding bonds at t .

A key market-composition object is the peril-category weight:

$$w_{c,t} = \frac{\sum_{i=1}^{N_{c,t}} Size_{i,t}}{\sum_{j=1}^{N_t} Size_{j,t}}, \quad (13)$$

where $N_{c,t}$ is the number of outstanding bonds in peril c at time t .

5.5 Interpreting Table 2: what drives beta dispersion

Table 2 runs pooled regressions (bond $\times$ quarter) to explain $\hat{\beta}_{i,t}$ with its building blocks and provides univariate R^2 diagnostics.

Column (1): a “kitchen-sink” linear approximation. With bond characteristics plus peril and time fixed effects, the linear regression achieves $R^2 \approx 0.83$. This shows that although $\hat{\beta}$ is generated by a nonlinear simulation, it is well-approximated by a linear function of observables.

Univariate R^2 column: which inputs matter most? Discount margin (dm) has very high univariate explanatory power ($R^2 \approx 0.57$), and actuarial probabilities (ap , el , ep) are each very informative ($R^2 \approx 0.44$ – 0.47). Peril fixed effects also matter ($R^2 \approx 0.48$), consistent with systematicity depending on peril composition.

Columns (3)–(5): a parsimonious representation. The paper emphasizes $el_{i,t}$ (“quantity of expected losses”) and $w_{c,t}$ (“systematicity via peril concentration”):

- Using (el, w) alone yields $R^2 \approx 0.61$ (column (3)).
- Adding an interaction $el \times w$ barely changes R^2 but the interaction itself is highly informative (column (4)).
- The interaction alone achieves $R^2 \approx 0.50$ (column (5)), making it the single strongest parsimonious determinant.

Table 2
Determinants of $\hat{\beta}$

	(1)	(2)	(3)	(4)	(5)	Univariate R^2
ap	-0.04** (0.02)					.45
el	0.15*** (0.05)		0.25*** (0.00)	.36*** (0.01)		0.47
ep	0.02 (0.04)					.44
$\ln(Size)$	0.04*** (0.01)					.06
dm	0.08*** (0.00)					.57
ref	0.07*** (0.02)					.00
w_c			1.77*** (0.04)	2.24*** (0.06)		.23
$el \times w_c$				-0.25*** (0.03)	0.61*** (0.01)	.50
$\hat{\beta}_{loss}$		1.07*** (0.00)				.99
Peril FE	Yes	No	No	No	No	.48
Time FE	Yes	No	No	No	No	.01
R^2	.83	.99	.61	.61	.50	
N	4,548	4,548	4,548	4,548	4,548	

The table presents estimation results from bond-quarter-level pooled regressions where the simulated betas ($\hat{\beta}_{i,t}$) are explained with its various determinants: Attachment probability (ap_i), expected loss (el_i), exhaustion probability (ep_i), natural logarithm of the issuance size ($\ln(Size_i)$), discount margin ($dm_{i,t}$), reference rate ($ref_{i,t}$), weight of bond i 's peril category in cat market portfolio ($w_{c,t}$), and an alternative version of simulated beta where discount margins of all bonds are set to zero ($\hat{\beta}_{loss}$). Standard errors are two-way clustered by bond and quarter. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 3
Pricing of catastrophe market risk

t	(1)			R^2	(2)			N
	$\lambda_{0,t}$	$\lambda_{cat,t}$	$\lambda_{cat,t}^0$		$\lambda_{cat,t}^0$	$\lambda_{cat,t}^{0,-}$	R^2	
2003	1.52	2.03	-1.53	0.74	3.09	-0.47	.94	44
2004	0.45	1.73	-0.59	0.93	1.93	-0.39	.96	55
2005	0.61	1.82	-0.75	0.94	2.11	-0.47	.97	55
2006	2.50	4.81	-1.47	0.59	5.98	-0.30	.81	44
2007	1.33	4.08	-1.09	0.88	4.74	-0.43	.94	70
2008	2.00	2.75	-1.77	0.82	3.85	-0.67	.91	67
2009	4.33	4.73	-4.07	0.83	7.40	-1.40	.91	58
2010	3.37	2.99	-3.07	0.71	5.15	-0.92	.91	65
2011	2.03	3.18	-1.93	0.83	4.41	-0.70	.94	55
2012	2.23	4.42	-2.33	0.78	5.91	-0.84	.95	84
2013	0.68	3.08	-0.75	0.76	3.54	-0.29	.94	88
2014	1.00	1.90	-0.96	0.68	2.57	-0.30	.91	100
2015	1.20	1.62	-1.19	0.56	2.40	-0.41	.86	100
2016	1.07	1.32	-0.92	0.63	1.97	-0.27	.88	92
2017	0.75	1.25	-0.58	0.47	1.72	-0.11	.83	122
2018	0.53	1.33	-0.37	0.52	1.65	-0.05	.83	122
FM	1.62*** (0.29)	2.37*** (0.25)	-1.51*** (0.27)	0.66	3.32*** (0.38)	-0.56*** (0.10)	0.87	63

The table presents estimation results for cross-sectional regressions of the form

$$E_t(R_{i,t+1}^e) = \lambda_{0,t} + \lambda_{cat,t} \hat{\beta}_{i,t} + \varepsilon_{i,t},$$

Column (2): robustness to an alternative “loss-only” beta. $\hat{\beta}^{loss}$ is computed by setting discount margins to zero (isolating disaster cash-flow variation). Regressing $\hat{\beta}^{loss}$ on $\hat{\beta}$ yields a slope about 1.07 and $R^2 \approx 0.99$, showing that most beta dispersion is driven by downside loss states rather than upside coupon variation.

6 Main evidence: “failure to share” disaster risk

6.1 Cross-sectional pricing test (Equation (14) and Table 3)

The core test is a (quarterly) Fama–MacBeth cross-sectional regression:

$$E_t(R_{i,t+1}^e) = \lambda_{0,t} + \lambda_{cat,t} \hat{\beta}_{i,t} + \varepsilon_{i,t}. \quad (14)$$

Interpretation of Table 3 (year-by-year and average pricing).

- $\lambda_{cat,t}$ is **positive in every year** (2003–2018), ranging from about 1.25% to 4.81% in the June cross-sections.

- Cross-sectional fit is very high: R^2 ranges from about 0.47 to 0.94 year-by-year.
- The Fama–MacBeth average price of cat risk is about 2.37% (Newey–West s.e. ≈ 0.25), strongly rejecting the frictionless prediction of zero pricing.
- Comparing $\lambda_{cat,t}$ to the *theoretical* benchmark $E_t(R_{cat}^e)$, the empirical slope is smaller, consistent with attenuation from beta measurement error and an imperfect proxy for specialists’ true wealth portfolio.

6.2 Figure 1: security-level fit of the intermediary model

Figure 1 plots bond-level expected excess returns ($dm - el$) against predicted values from the one-factor intermediary model. The tight positive relationship visually supports the claim that the intermediary-marginal-utility proxy prices the cross-section; the paper reports that about **71%** of security-level expected-return variation is explained.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
λ_0	1.62*** (0.29)	2.52*** (0.40)	1.92*** (0.34)	0.44 (0.28)	2.48*** (0.38)	1.21*** (0.30)	1.79*** (0.30)
λ_{cat}	2.37*** (0.23)					1.80*** (0.21)	1.67*** (0.17)
λ_{el}		1.00*** (0.13)		0.99*** (0.13)		0.29*** (0.08)	
λ_w			8.00*** (1.54)	6.74*** (1.18)		1.87* (1.00)	
$\lambda_{el \times w}$					3.44*** (0.56)		0.99*** (0.27)
N	63	63	63	63	63	63	63
R^2	.66	.47	.19	.63	.62	.70	.69

The table presents estimation results for quarterly Fama–MacBeth regressions of the form

$$E_t(R_{i,t+1}^e) = \lambda_{0,t} + \lambda_{cat,t} \hat{\beta}_{i,t} + \lambda_{el,t} el_i + \lambda_{w,t} w_{c,t} + \varepsilon_{i,t},$$

where the dependent variable is the expected excess return on cat bond i in the end of quarter t . $\hat{\beta}_{i,t}$ is the simulated beta estimate, el_i expected loss, and $w_{c,t}$ is the weight of bond i ’s peril category in cat market portfolio. R^2 is the time-series average of cross-sectional figures. Newey and West (1987) standard errors with four lags are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

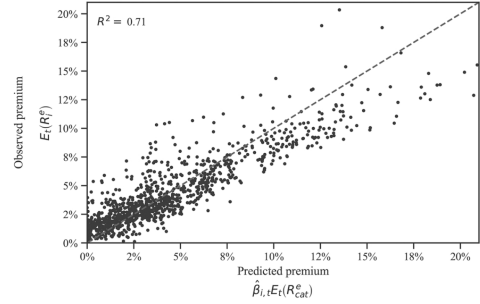


Figure 1
Observed versus predicted risk premium
The figure plots expected excess returns (discount margin - expected actuarial loss) on sample cat bonds between 2003 and 2018 against their predicted values from a single-factor intermediary model. Bonds’ cat market betas are estimated from 500,000 years of simulated disaster data, and each dot represents a single cat bond observation in the end of June of a given year.

6.3 Characteristics vs factor exposure (Table 4)

Table 4 asks whether pricing is “really beta” or merely characteristics. Key takeaways:

- Expected loss (el) and peril weight ($w_{c,t}$) each price the cross-section on their own, and jointly explain about 63%.
- The one-factor beta model performs similarly ($R^2 \approx 0.66$).
- The interaction $el \times w$ is especially strong, consistent with the idea that the combination of “riskiness” and “systematicity” matters for specialist exposure.
- Overall, the simulation-based beta provides a theory-consistent way to compress multiple characteristics into a single systematic-risk index.

7 Time-series and pooled evidence on intermediary capital

7.1 CRRA mapping (Equation (15)) and price of risk (Equation (16))

To connect mean–variance pricing to a standard CRRA preference, the paper uses:

$$U(w_m) = \frac{w_m^{1-\hat{\rho}} - 1}{1 - \hat{\rho}}. \quad (15)$$

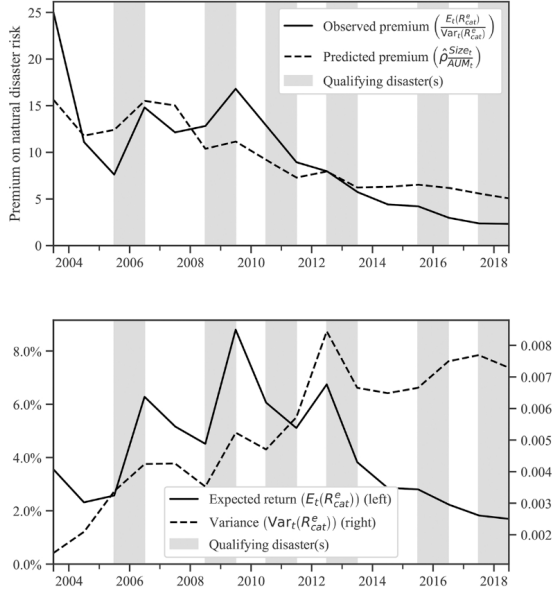


Figure 3
Time-series evolution of price of natural disaster risk
 Panel A plots observed and predicted premium on natural disaster risk, where coefficient of relative risk aversion $\hat{\rho} = 4.6$. $R_{cat,t}^e$ is return on cat market factor portfolio, $Size_t$ is the size of the market, and AUM_t is total assets under management of the specialist funds in year t . Shaded regions represent years (starting in July) during which qualifying loss events occurred. Panel B plots the two components of observed premium separately.

Table 5
Pricing of catastrophe market risk (pooled regressions)

A. $\beta_{i,t} = \hat{\beta}_{i,t}$			
F_t	$\text{Var}_t(R_{cat,t+1}^e) \frac{Size_t}{AUM_t}$	$\frac{Size_t}{AUM_t}$	$E_t(R_{cat,t+1}^e)$
$F_t \beta_{i,t}$	3.54*** (0.15)	0.02*** (0.00)	0.88*** (0.03)
R^2	.81	.75	.90
N	1,221	1,221	1,221
B. $\beta_{i,t} = \phi_0 + \phi_1 el_i + \phi_2 w_{c,t}$			
F_t	$\text{Var}_t(R_{cat,t+1}^e) \frac{Size_t}{AUM_t}$	$\frac{Size_t}{AUM_t}$	$E_t(R_{cat,t+1}^e)$
F_t	1.74*** (0.32)	0.00 (0.00)	0.28*** (0.05)
$F_t el_i$	0.68*** (0.13)	0.00*** (0.00)	0.21*** (0.02)
$F_t w_{c,t}$	3.41*** (0.79)	0.04*** (0.00)	1.13*** (0.13)
R^2	.76	.83	.90
N	1,221	1,221	1,221

The table presents estimation results for pooled regressions of the form

$$E_t(R_{i,t+1}^e) = \delta F_t \beta_{i,t} + \varepsilon_{i,t},$$

where the dependent variable is the expected excess return on cat bond i in the end of June of year t . Standard errors clustered by issuer are shown in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

Locally, CRRA implies mean–variance risk tolerance $\rho = \hat{\rho}/w_m$. Substituting into (10) yields a time-series restriction:

$$\frac{E_t(R_{cat,t+1}^e)}{\text{Var}_t(R_{cat,t+1}^e)} = \hat{\rho} \frac{Size_t}{AUM_t}, \quad (16)$$

where $Size_t$ is market size and AUM_t is specialists' total assets under management.

Interpretation. The premium per unit of risk is high when specialist capital is scarce relative to market size, and falls as outside capital access improves.

7.2 Figure 3: aggregate premium weakens as capital deepens

Figure 3 compares observed and predicted premium per unit of risk over time. The key pattern is a downward trend after 2010 and a weaker premium spike after later disasters (relative to early sample years), consistent with an expansion in specialist-fund capital (or a loosening of the effective α constraint).

7.3 Pooled restriction and estimation (Equations (17)–(18); Table 5)

Equations (9) and (16) imply:

$$E_t(R_{i,t+1}^e) = \hat{\rho} \text{Var}_t(R_{cat,t+1}^e) \frac{Size_t}{AUM_t} \beta_{i,t}. \quad (17)$$

The paper estimates pooled regressions:

$$E_t(R_{i,t+1}^e) = \delta F_t \beta_{i,t} + \varepsilon_{i,t}, \quad F_t \equiv \text{Var}_t(R_{cat,t+1}^e) \frac{Size_t}{AUM_t}, \quad (18)$$

with alternative F_t definitions and two beta specifications (simulated vs characteristics-based).

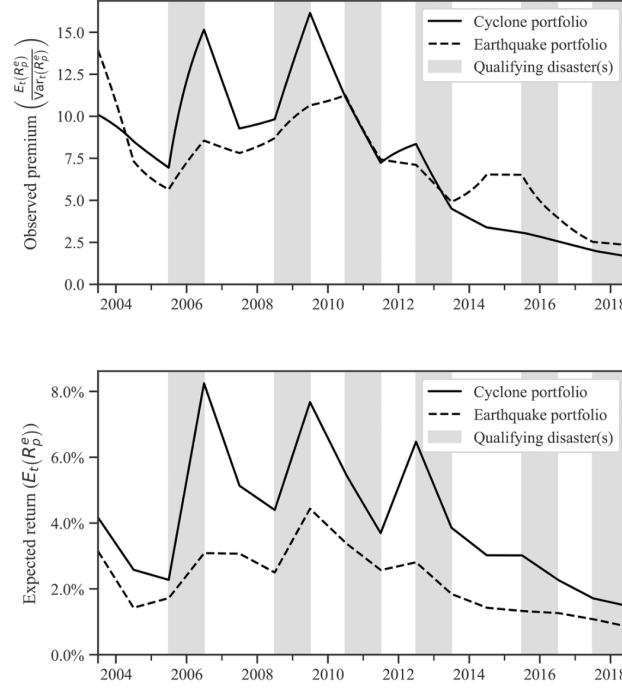


Figure 5
Time-series evolution of price of cyclone and earthquake risks
 Panel A plots observed premium on portfolios of catastrophe bonds that are only exposed to cyclone and earthquake risks, respectively. Shaded regions represent years (starting in July) during which qualifying loss events occurred. Panel B plots the expected returns on these portfolios.

Interpretation of Table 5.

- **High explanatory power.** Across specifications, pooled R^2 is large (roughly 0.75–0.90), indicating strong joint fit of cross-section and time-series variation.
- **Economic meaning of δ .** In the main specification (Panel A), the coefficient on $F_t\beta_{i,t}$ is about 3.54. With the paper’s mapping, this can be interpreted as a proxy for specialists’ effective risk aversion.
- **Robustness to beta representation.** Panel B shows that using a characteristics-based beta ($\beta_{i,t} = \phi_0 + \phi_1 el_{i,t} + \phi_2 w_{c,t}$) preserves strong fit.

8 Testing the exclusion restriction: disasters and macro aggregates

8.1 Disaster damage measure (Equation (19))

Using international disaster data, define country-level disaster damages relative to GDP:

$$dmg_{i,t} = \frac{\sum_d DMG_{d,i,t}}{GDP_{i,t-1}}. \quad (19)$$

8.2 Local projections (Equation (20)) and Figure 4

Impulse responses are estimated via Jordà (2005) local projections:

$$\Delta^h y_{i,t+h} = \gamma_i + \gamma_t + b_1 \text{Small}_{i,t} + b_2 \text{Large}_{i,t} + \varepsilon_{i,t}, \quad (20)$$

where $\Delta^h y_{i,t+h} \equiv y_{i,t+h} - y_{i,t-1}$, $\text{Small}_{i,t} = \mathbb{1}\{0.1\% \leq dmgi_{i,t} \leq 1\%\}$, and $\text{Large}_{i,t} = \mathbb{1}\{dmgi_{i,t} > 1\%\}$.

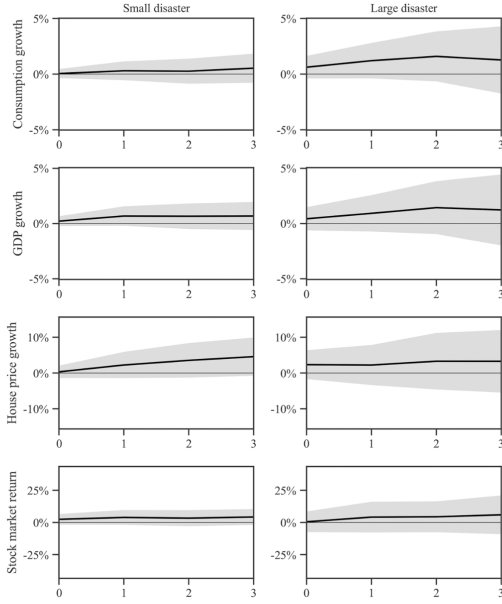


Figure 4
Impulse responses of selected macroeconomic variables to natural disasters occurring in year 0
Shaded region shows 95% confidence interval on the estimates. Standard errors are clustered by year. Sample includes 13 developed countries from 1950 to 2020.

Table 6
Catastrophe market risk and liquidity

A. Correlation matrix							
	$\hat{\beta}$	Turnover	N trades	Trd intvl	Size	Age	Maturity
$\hat{\beta}$	1.00	.01	-.08	.05	-.20	-.14	-.04
Turnover	.01	1.00	.64	-.11	.04	-.14	.14
N trades	-.08	.64	1.00	-.24	.42	-.11	.14
Trd intvl	.05	-.11	-.24	1.00	-.20	.30	-.19
Size	-.20	.04	.42	-.20	1.00	-.02	.09
Age	-.14	-.14	-.11	.30	-.02	1.00	-.60
Maturity	-.04	.14	.14	-.19	.09	-.60	1.00
DS	.06	.00	-.06	-.00	-.11	-.03	-.05
B. Pricing of catastrophe market risk and liquidity							
LIQ	Turnover	N trades	Trd intvl	Size	Age	Maturity	DS
λ_0	1.64*** (0.30)	1.60*** (0.29)	1.54*** (0.26)	1.66*** (0.28)	1.64*** (0.30)	1.65*** (0.30)	1.93*** (0.30)
λ_{cat}	2.44*** (0.28)	2.42*** (0.27)	2.42*** (0.31)	2.39*** (0.28)	2.43*** (0.28)	2.44*** (0.28)	2.07*** (0.35)
λ_{liq}	-0.10** (0.04)	-0.27*** (0.08)	-0.02 (0.04)	-0.30*** (0.11)	-0.16** (0.07)	0.03 (0.08)	1.08** (0.45)
N	57	57	57	57	57	57	41
R ²	.66	.66	.66	.67	.67	.67	.62

Panel A presents correlations of simulated cat market betas ($\hat{\beta}$) and various liquidity measures of. Annualized turnover, number of quarterly trades, trading interval (number of days since the bond last traded), and dealer spread (DS, the quarterly average of the difference between volume-weighted dealer buy and sell prices in a given trading day) are measured using data from TRACE. Characteristics-based measures include size (amount issued), age, and time to maturity. All liquidity measures are scaled by their pooled standard deviation. Panel B presents estimation results for quarterly Fama-MacBeth regressions of the form

$$E_t(R_{i,t+1}^e) = \lambda_{0,t} + \lambda_{cat,t} \hat{\beta}_{i,t} + \lambda_{liq,t} LIQ_{i,t} + \varepsilon_{i,t},$$

Interpretation of Figure 4. Across macro variables (consumption growth, GDP growth, house prices, stock market returns), impulse responses to both small and large disasters are close to zero. This supports the maintained assumption that, at the macro level relevant for household MRS, disasters are approximately orthogonal to aggregate economic conditions.

9 Ruling out alternative explanations

9.1 Liquidity (Table 6)

A natural competing explanation is a liquidity premium. Table 6 provides two key facts:

- **Low correlation between cat betas and liquidity proxies.** In Panel A, $\hat{\beta}$ has near-zero correlations with trading-based measures (turnover, number of trades, trading interval) and with dealer spreads.
- **Cat-risk price is robust to controlling for liquidity.** In Panel B, the estimated λ_{cat} remains positive and significant across liquidity controls, with very similar R^2 .

Conclusion from Table 6: cross-sectional liquidity differences do not appear to be the main driver of the beta-pricing relation.

9.2 Climate change concerns (Figure 5)

If cyclone risk were tied to systematic climate risk, cyclone bonds might load on macro risks, violating the identification restriction. Figure 5 compares cyclone-only and earthquake-only portfolios:

- Premium per unit of risk falls in parallel for both cyclones and earthquakes through the 2010s.
- The similarity suggests that any additional “climate-change premium” in cyclone bonds is not large enough to dominate aggregate pricing patterns in this sample.

9.3 Other explanations

The paper discusses further possibilities in the Internet Appendix (e.g., peso problems, probability weighting). A central idea is that many such alternatives would imply pricing patterns that can be differenced out (e.g., comparing tranches within the same issuance), while the main beta-pricing relation persists.

10 Short summary

Cat bonds provide a rare laboratory in which the key omitted-variable critique to intermediary asset pricing is muted: natural disasters are plausibly orthogonal to aggregate wealth. In this setting, a theoretically motivated proxy for specialist intermediaries’ marginal utility prices most of the cross-section of expected returns and helps explain time-series movements in the catastrophe risk premium. The evidence supports the view that **limited intermediary capital and segmented participation can generate a large, time-varying premium even for risks that are diversifiable in aggregate.**

11 Conclusion

11.1 Summary

This paper concludes that **catastrophe bond (cat bond) prices reveal a meaningful failure of risk sharing in markets for natural-disaster risk.** Although natural disasters are largely *idiosyncratic at the macroeconomic level*, cat bond investors earn a substantial and systematically priced risk premium that is best explained by **market segmentation and intermediary constraints**, rather than by standard frictionless, representative-agent asset-pricing logic.

The argument is internally consistent and proceeds in one chain:

1. **Identification advantage: disasters are (approximately) macro-idiosyncratic.** The paper motivates and empirically supports the assumption that natural disasters do not have a first-order effect on aggregate consumption, GDP growth, house prices, or stock market returns using a long macro panel (13 developed countries, 1950–2020) and local projections.
2. **Yet cat bonds earn large, heterogeneous expected excess returns.** Expected excess returns are measured using a bond-level actuarial object:

$$\mathbb{E}_t[Re_{i,t+1}] \equiv \underbrace{dm_{i,t}}_{\text{discount margin}} - \underbrace{\mathbb{E}_t[\ell_{i,t+1}]}_{\text{expected actuarial loss}},$$

where the expected loss is informed by catastrophe models used by market participants.

3. **A one-factor intermediary model explains a large share of the cross section.** Using simulated (forward-looking) “cat market betas” that measure exposure to disaster states when specialists are marginal, the paper shows that a simple single-factor pricing relation

$$\mathbb{E}_t[Re_{i,t+1}] = \lambda_{0,t} + \lambda_{\text{cat},t} \beta_{i,t} + \varepsilon_{i,t}$$

has very high explanatory power. In the headline summary, **about** 71% of the security-level variation in expected excess returns is explained by the model-implied intermediary pricing kernel (as emphasized in the abstract and illustrated by the observed-versus-predicted plot).

4. **The time series of the aggregate price of disaster risk moves with intermediaries’ ability to raise capital.** In the intermediary framework, the *price per unit of risk* declines when specialist capital becomes abundant relative to market size:

$$\frac{\mathbb{E}_t[Re_{\text{cat},t+1}]}{\text{Var}_t(Re_{\text{cat},t+1})} = \hat{\rho} \frac{\text{Size}_t}{\text{AUM}_t},$$

where AUM_t is the capital managed by specialist cat-bond funds and Size_t is the size of the catastrophe-risk market. Empirically, the aggregate premium declines and becomes less sensitive to the occurrence of disasters when AUM_t rises, consistent with relaxed intermediary constraints.

5. **Alternative explanations are unlikely to be the main drivers.** The paper considers and tests several competing stories (including liquidity/transaction-cost premia, climate-change pricing, a “peso problem” driven by rare unobserved mega-disasters, and probability-weighting style explanations), and finds that these alternatives do not overturn the central intermediary/segmentation interpretation.
6. **Interpretation: a failure of risk sharing.** Because macroeconomic diversification is plausible for these disaster risks, frictionless models would predict a much smaller compensation for risk. The fact that disaster-risk premia are large and move with specialist capital is therefore consistent with **segmented pricing** where **asset-class-specific risks are priced** even if they would wash out in the aggregate.

11.2 Why “failure to share” shows up as a premium

The paper’s intuition is a direct application of intermediary asset pricing to an asset class whose fundamental risks are not macro-driven.

1) What would happen in frictionless, well-shared risk markets? If households (or a representative investor) could costlessly absorb natural-disaster risk, and if disasters are truly idiosyncratic at the macro level, then catastrophe risk should be close to **actuarially fairly priced**: the required expected excess return beyond expected loss should be small.

2) What changes when markets are segmented and specialists are marginal? In practice, the cat bond investor base is specialized. If the marginal investor is a catastrophe-risk specialist (e.g., dedicated ILS fund managers), then prices reflect the **specialists’ marginal utility**, not the marginal utility of the diversified household sector. After disasters, specialists suffer losses and become more constrained; their marginal utility rises. Assets that load more on “specialist bad times” require higher expected returns.

3) Why betas matter here. A bond’s “cat market beta” measures how strongly it loses value in the same disaster states that hurt the specialist sector. Higher beta means the bond is more exposed to intermediary distress states, so it earns a higher expected excess return.

4) Why the premium falls when outside capital becomes more available. When specialists can attract more outside capital (higher AUM_t) relative to market size, their constraint relaxes. Then each unit of disaster risk requires less compensation:

$$\begin{aligned} \text{More specialist capital} &\Rightarrow \text{looser constraint} \Rightarrow \text{lower price of disaster risk} \\ &\Rightarrow \text{weaker post-disaster price pressure.} \end{aligned}$$

This is the “slow-moving/limited-entry capital” logic applied to catastrophe risk.

5) A microfoundation for the barrier to entry. The paper emphasizes a practical mechanism consistent with segmentation: **high fixed costs of learning and infrastructure**. Institutional investors are often unfamiliar with cat bond structure and catastrophe-modeling, and entering the market requires data, expertise, and operational capability. Given the market’s relatively small size, these fixed costs deter participation, preventing capital from arbitraging away the premium.

11.3 Key quantitative takeaways

- **Cross-section fit:** the one-factor intermediary model explains a very large share of expected-return dispersion (headline: about 71% of security-level variation in expected excess returns).
- **Risk-aversion calibration:** a back-of-the-envelope mapping implies a specialist relative risk aversion in the range $\hat{\rho} \approx 4.6$ to 9.2 depending on assumptions about specialists’ market share.
- **Time-series implication:** the aggregate premium decreases and becomes less sensitive to disasters when specialists’ access to outside capital improves (higher AUM_t relative to $Size_t$).

11.4 Takeaway

Tomunen (2026) shows that catastrophe bonds exhibit large, systematically priced disaster-risk premia that are best explained by a segmented-market, intermediary-constraint framework—even though natural disasters appear largely macro-idiosyncratic—implying a failure of risk sharing driven by limited specialist capital and high fixed costs of market entry.